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Thermal Management of Solar Photovoltaic Panels: Comparative Analysis of Water and Ethylene Glycol Cooling

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ABSTRACT

Solar photovoltaic (PV) systems are increasingly being adopted as a key renewable energy source to mitigate carbon emissions from electricity generation. However, it is well recognized that elevated temperatures significantly reduce the efficiency and performance of PV panels. To improve panel performance, a coolant consisting of water and ethylene glycol is used. In this paper, an experimental investigation is carried out using water and a mixture of 10% ethylene glycol with water, stored in a rectangular container attached directly to the backside of the panels. The novelty of this study lies in the use of a water and 10% ethylene glycol mixture as a hybrid coolant. This mixture provides better heat transfer, temperature uniformity, and thermal stability than pure water, offering a cost-effective and practical solution for PV cooling in hot climates. A comparative analysis of PV panels with and without cooling is conducted. Additionally, a mathematical model of heat transfer for the cooling system is presented. The results show that using water alone and using a 10% ethylene glycol solution mixed with water as the cooling fluid reduces PV panel temperatures by up to 9.75% and 15.66%, respectively. This leads to a maximum increase in electrical efficiency of 9.42% and 11.01%, respectively. For the water-cooled panel and the panel cooled with 10% ethylene glycol mixed with water, the 40 W panel's maximum power output increases by 5.043 W and 6.63 W when solar intensity is at its peak.

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1. INTRODUCTION

With advancements in next-generation energy technologies, solar photovoltaic (PV) power generation has attracted considerable research interest. According to the International Energy Agency, approximately 190 GW of PV power plants were projected to be commissioned globally in 2022, marking a 25% increase from 2021 (International Energy Agency, 2023). The study shows that temperature significantly impacts the efficiency of solar PV panels, primarily through its adverse effect on the open-circuit voltage (VOC). Under full sunlight, the surface temperature of PV panels can exceed ambient conditions by about 40°C, with operating temperatures ranging between 60–80°C, limiting power generation efficiency to around 10% (Biwole et al., 2013; Patil et al., 2023). Several studies report that for crystalline silicon PV cells, efficiency decreases by 0.4–0.65% for every 1°C increase above 25 °C (Bhakre et al., 2021; Biwole et al., 2011; Teng et al., 2024). Keeping the panel temperature close to the ambient air temperature is a major concern among researchers. Thus, cooling of solar panels is widely recognized as an effective approach to enhancing energy conversion efficiency (Bhakre et al., 2021; Okta et al., 2024; Sato & Yamada, 2019).

Various techniques are employed, including natural and forced methods using air or water as working fluids (Harmailil et al., 2024), evaporative cooling (Bhakre et al., 2023a), heat pipes or microchannels as heat exchangers (Ma et al., 2024), water immersion (Ibrahim et al., 2024), phase change materials (Bhakre et al., 2022), and PVT (Bhakre et al., 2023b; Korkut et al., 2022). Many of these cooling strategies show significant improvements in solar PV panel performance, whereas some involve complex designs that are less feasible in regions with limited technological capabilities. Passive cooling, relying on natural heat dissipation, offers modest efficiency improvements without additional energy input, whereas active cooling employs external power to circulate working fluids, achieving more substantial results (Sharaf et al., 2022). Therefore, cooling PV panels is a proven method to enhance their efficiency. Among various cooling technologies, liquid-based cooling systems, which use water along with nanoparticles for thermal regulation, have demonstrated significant cooling effects and are considered one of the most effective approaches (Abdo et al., 2020; Abou et al., 2025; Chanphavong et al., 2022).

Over the past two decades, water-cooled PV panels have been extensively studied by both experimental and numerical methods due to the high thermal conductivity and widespread availability of water. Abdo et al. (2020) presented seawater as a cooling fluid, mixed with saturated activated alumina using six different salinities of water. The results show that saturated activated alumina tablets with saline water improved cooling performance by 3–4°C compared to pure water. Hadipour et al.

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(2021) designed a novel pulsed-spray water cooling system capable of reducing water consumption by one-ninth at 0.2 duty cycles. Their results show that maximum electrical power output increased by 25.9% relative to the reference panel. Shastry & Arunachala (2020) introduced a novel aluminum metal matrix with a honeycomb structure in which PCM is filled, which can be effectively used in PV and PV/T systems, enhancing diffusion rates compared to solid metal plates. Outdoor experiments revealed significant improvements in various energy parameters, including conversion efficiency, with a maximum improvement of 8.6% in PCM-based PV/T achieved alongside an 11.1% temperature drop. Arunkumar et al. (2023) experimentally compared several cooling methods, such as heat sink, water jacket, air bubble cooling, and combinations of these techniques. Sending air bubbles into the cooling jacket creates turbulent motion, which cools the PV panel efficiently. With a maximum panel temperature drop of 15.36%, the combination of heat sink, air bubble, and water jacket produced the greatest cooling percentage. Homadi et al. (2020) introduced a novel cooling technique named the thermos-magneto-generator (TMG), which is coupled to the back of solar panels, functioning as both a heat dissipator and an electrical generator. The system covers two regions: one being the heat source (the solar panel), and the other acting as the heat sink (a flat plate at atmospheric temperature). Experimental and theoretical testing using COMSOL Multiphysics software was conducted, varying solar radiation from 500 W/m² to 1000 W/m². Jasim (2022) experimentally studied water spray cooling factors such as angle, distance, and number of nozzles. The combination with a 20° spray angle gave the best results, increasing electrical efficiency to 18.76% and lowering the panel temperature from 60°C to 26°C. The best cooling occurred at lower spray heights and with fine mist setups. Basem et al. (2024) compared four cooling methods and found that spraying water on the front surface was the most effective. It increased panel efficiency to 22% and kept the surface temperature at 41°C. The study showed that water provides better cooling uniformity and is simpler than air and phase-change methods. Mostakim et al. (2024) developed a new front-surface water-spray PV/T design that combines electricity generation and hot water production. Spraying water on the panel's surface improved electrical efficiency and thermal recovery. It also prevented dust buildup.

With development in analysis tools and computational power, numerical studies are gaining more attention due to cost-effectiveness, time efficiency, and the ability to optimize designs in the early stages of development. Yildirim et al. (2022) designed a cooling box for PV panels to optimize thermal and electrical power generation efficiency. The simulation results show an improved electrical efficiency of 17.79% and a thermal efficiency of 76.13% at an inlet water temperature of 15°C and a mass flow rate of 0.014 kg/s. Yang et al. (2024) proposed a new cooling method called dew point evaporative cooling (DPEC) with two wet channels. One channel is for the cooler (conventional DPEC), which produces a pre-cooled air supply, and the other channel is attached to the back of the PV panel. The proposed method consistently gave greater than 15% power generation efficiency under two different summer conditions. Mankani et al. (2022) proposed a computational fluid dynamics (CFD) analysis of a solar PV/T system with a bottom active air-cooling system. The study model was developed using Fluent (ANSYS) software, combining a bottom air duct with an inlet/outlet manifold for uniform airflow to the backside of the PV/T system. A maximum outlet temperature of 48.8°C was obtained under solar radiation of 800 W/m² at a mass flow rate of air of 0.01 kg/s.

In the early 21st century, alongside advancements in technology, nanomaterials composed of nanoscale particles were developed and gained significant attention in the solar energy sector. Their advantageous properties also made them suitable for integration with cooling agents in various studies, enhancing heat removal and storage capabilities. Salehi et al. (2023) introduced the use of aluminum nanoparticles to enhance the cooling performance of the fluid. Their findings show that solar panel efficiency and output power increased by an average of 13.5% and 13.7%, respectively, compared to water cooling without nanoparticles. Suja et al. (2023) proposed swirling nanofluid jets impinging as a novel method for cooling high-concentrator solar cells. Numerical analysis revealed that electrical efficiency increased by 39.43% with Al_2O_3-water at 50 g/min and 39% with CuO-water at 25 g/min. Al-Waeli et al. (2019) studied how to improve the performance of PV/T systems using a combined nanofluid (SiC-water) and nano-PCM cooling method. They developed and compared three mathematical prediction models against results from an artificial neural network (ANN). Their work showed a strong correlation ($R^2 = 0.81$) and low error values (MSE = 0.0361, RMSE = 0.371). Their findings indicated that using nanofluid and nano-PCM cooling greatly improves heat dissipation and the overall efficiency of PV/T systems while also lowering the prediction error in performance estimation. Kazem et al. (2021) conducted a study on a 100 W web-configured PV/T system in Sohar, Oman, using water and SiC–water nanofluid as coolants. The results show that electrical efficiency increased by up to 20.21%, and thermal efficiency rose by 43.3% when using the nanofluid. The study also pointed out that nanofluid-based PV/T systems work well under hot conditions. Ebaid et al. (2024) compared $CaCO_3$ (metallic) and Fe_3O_4 (non-metallic) nanofluid coolants in PV panels at different flow rates under outdoor conditions. The results showed that both nanofluids significantly improved heat transfer and electrical power output compared to pure water systems, especially under turbulent flow conditions. Jose et al. (2023) conducted a CFD and experimental study with Al_2O_3-water nanofluid (0.1%, 0.2%) in serpentine copper tubing, which showed improved overall PV efficiency. This improvement came from better temperature uniformity and higher convective performance. The research confirms that the coolant flow rate is a key design factor. The study carried out by Chaichan et al. (2023) shows that ethylene glycol works better than water as a coolant because it has higher viscosity, lower vapor pressure, and better thermal stability. In another study, Aselebe et al. (2024) examined the thermal behavior of a hybrid nanofluid based on ethylene glycol. This fluid contains CuO and Al_2O_3 nanoparticles and is used over a stretchable or shrinking flat plate for cooling solar panels. The study showed that ethylene glycol, because of its higher viscosity and specific heat capacity, effectively stabilizes temperature changes and improves heat absorption.

In the literature review, most studies are carried out with water and water-nanofluid combinations. These studies show significant improvement in the thermal management of solar panels. However, although these active cooling systems provide better cooling performance, their main drawback is the operational expense, as they require pumps powered by electricity. In contrast, passive cooling uses phase change materials (PCMs) as the cooling fluid but is limited by higher melting temperatures. The absence of a suitable cooling fluid

in passive cooling leads to lower electrical efficiency. The present study focuses on a stationary container; nanoparticles tend to settle over time because of gravity and the lack of fluid movement. This settling causes uneven particle distribution and lowers heat transfer efficiency. Since this study looks at a stationary container, nanoparticle sedimentation is important. It may reduce the long-term stability and thermal conductivity of the nanofluid. Therefore, even though nanofluids can initially improve heat transfer, maintaining a uniform suspension in static conditions is a big challenge for real-world thermal management. This study introduces a novel approach by using a mixture of water and 10% ethylene glycol as a hybrid coolant for heat management in photovoltaic systems. While most previous studies focus on pure water or nanofluids, this work presents a low-cost, readily available, and thermally stable mixture that enhances heat transfer without substantially increasing viscosity or complicating the system. The primary aim is to investigate the effect of the water-ethylene glycol mixture on the thermal performance of PV panels and compare it with water cooling and existing literature.

2. EXPERIMENTAL

2.1 Experimental setup

Figure 1 presents a simplified schematic diagram of the experimental setup used to evaluate the performance of photovoltaic (PV) panels under various cooling conditions. The setup consists of three configurations: (a) a non-cooled panel serving as the reference, (b) a water-cooled panel, and (c) a panel cooled with a 10% ethylene glycol mixed with water (EGMWW) solution. The panels are connected to a data storage unit for recording temperature, power output, and other parameters. A rheostat is incorporated into the circuit to control the electrical load on the panels, enabling consistent evaluation of power performance across all configurations. Multimeters are used to measure voltage and current, ensuring accurate calculation of power output.

Initially, the panels were tested under outdoor conditions to observe temperature fluctuations. Three thermocouples were attached to the backside of each panel, and measurements were taken over five days. The panel with the minimum temperature fluctuation was selected for the experimental study. A 40 W polycrystalline PV panel is used for the experimental study, and its various electrical properties under standard test conditions (STC) are shown in Table 1. An aluminum plate is used to create a heat sink container by fixing it to the backside of the PV panel. To fix the aluminum plate and thermocouples on the backside of the PV panel, and to ensure a leak-proof system, an epoxy solution is used. The fabricated panels with an aluminum plate for the experimental study are shown in Figure 2. The aluminum material has higher thermal conductivity, which allows more heat energy to pass through the aluminum plate and dissipate into the ambient air. Three RTD PT100 (3-wire, washer-type) thermocouples are placed at optimal distances on the backside of the PV panel to measure its temperature. All RTDs are fixed using high-temperature adhesive tape (Kapton tape). During the experiment, solar radiation was recorded with a pyranometer. The complete specifications of the measuring devices are provided in Table 2.

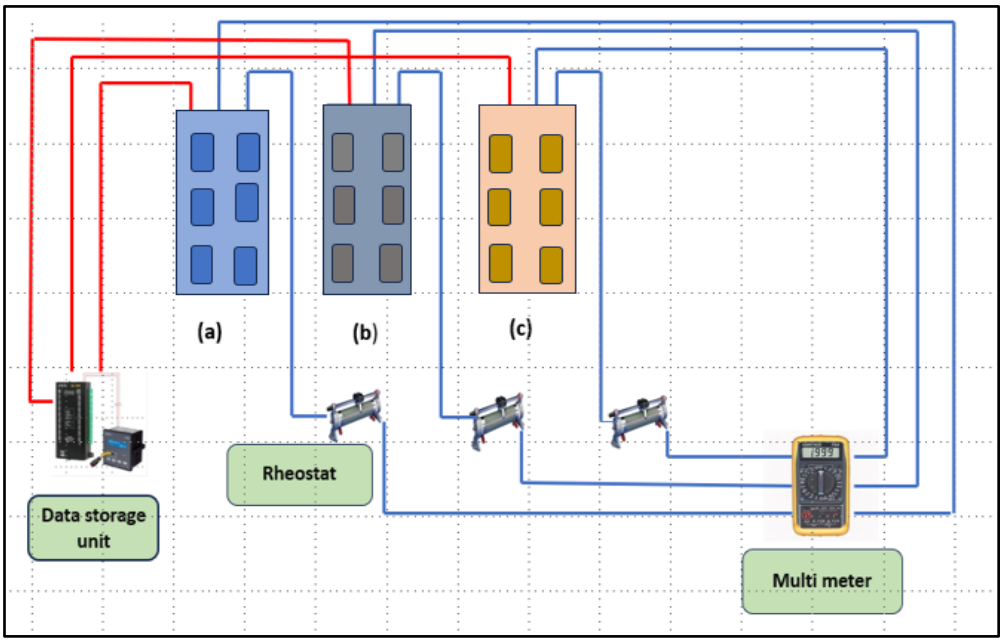


Figure 1. Simplistic diagram of experimental setup (a) non-cooled panel, (b) water-cooled panel, and (c) 10% ethylene glycol mixed with water-cooled panel.

Table 1. PV panel's properties at STC (25ºC, 1000 W/m^2).

Cell type	Polycrystalline
weight	3.3Kg
Dimensions(mm)	670×465×30
Module efficiency	13.48%
Maximum power output (P_{max}) W	40W
Maximum power current (I_{mp})	2.14A
Maximum power voltage (V_{mp})	18.70V
short circuit current (I_{sc})	2.25A
Open circuit voltage (V_{oc})	22.50V

Table 2. Specifications about the instruments/measurement apparatus used.

Sr. No.	Items	Brand	Model	Descriptions
1.	Pyranometer	Virtual electronics	Solar radiation recorder	range:0–2000W/m^2 Calibration:0.25 mV/W/m^2
2	Temperature sensor	PPI	RTD PT100	3-wire, washer-type Range: − 50 ºC to 250 ºC
3	Multi meter	Mastech	MAS830L	Range:0–10 DC Accuracy: standard: ±0.5 % of spa
4	Rheostat	TechDico	DICO	Range: 0–45 Ω, 3 A
5	Data storage unit	PPI	Unilog	16-channel data recorder Accuracy: ±0.25 % of reading

2.2 Cooling fluid

For the experiments, water is taken as the base cooling fluid because it has high specific heat (4186 J/kgK) and thermal conductivity (0.6 W/mK), meaning that it absorbs more heat per unit rise in temperature. Due to its high specific heat capacity, water can absorb and transfer a large amount of heat efficiently within the fluid domain. The second panel is filled with a cooling fluid consisting of 10% ethylene glycol mixed with water. Ethylene glycol is completely miscible in water and denser than water.

2.3 Leak-proof setup

During the first trial of the experiment, an aluminium plate was fixed using adhesive on the backside of the PV panel to create a heat sink container. The water-filled panel and the panel with 10% ethylene glycol mixed with water exhibited some leakage problems. The leakage occurred because, as the temperature of the water increased, its surface tension decreased. The reduced surface tension allowed the water to flow through very small holes. To resolve this issue, adhesive along with high-temperature glue was used to seal the leakage holes. The leakage problem was almost resolved.

2.4 2.4 Experimental procedure

The experimental setup is located on the rooftop of the Mechanical Engineering Department at VNIT, Nagpur, Maharashtra, India (21.1458°N, 79.0882°E). The solar panel is oriented facing south, as India lies in the Northern Hemisphere. To achieve the optimal solar angle, the tilt angle of the PV panel is adjusted by 15° for winter and/or summer, based on experimental studies conducted in Northern Hemisphere countries.

The actual setup, along with the measurement equipment used for the experimental installation, can be seen in Figure 3. The experiments were conducted over three consecutive days for each trial: from 29 February to 2 March and from 14 March to 16 March 2024, between 8:30 AM and 4:00 PM, under bright and sunny skies. Readings were taken at 1-minute intervals using a data storage unit. In this paper, results are presented for 15 March 2024.

2.5 Uncertainty analysis

The uncertainty analysis of the experimental data used the Coleman and Steele (1995) method to estimate the combined uncertainty in measured and derived parameters. This method takes into account both instrumental (systematic) and random (precision) errors linked to each measurement. The overall uncertainty in any derived quantity R, which depends on multiple measured variables x_1, x_2, ..., x_n, is expressed as:

$$\left(\frac{\delta R}{R}\right)^2 = \sum_{i=1}^{n}\left(\frac{\partial R}{\partial x_i}\cdot\frac{\delta x_i}{R}\right)^2 \quad (1)$$

For the photovoltaic system, the main parameters considered were power output and electrical efficiency.

The uncertainty of power output ($P = V \times I$) was determined using:

$$\frac{\delta P}{P} = \sqrt{\left(\frac{\delta V}{V}\right)^2 + \left(\frac{\delta I}{I}\right)^2} \quad (2)$$

The uncertainty of electrical efficiency $\eta_e = \frac{P}{G\times A}$ was obtained from:

$$\frac{\delta \eta_e}{\eta_e} = \sqrt{\left(\frac{\delta P}{P}\right)^2 + \left(\frac{\delta G}{G}\right)^2} \quad (3)$$

Based on these, the combined uncertainties were calculated as ±0.707% for power output and ±0.75% for electrical efficiency.

3. METHOD

3.1 Energy balance of the cooling system

The conservation of energy concept is used to calculate the rate of heating or cooling of the PV panel using Equation (4), shown in Figure 4.

$$\frac{dT_{PV}}{dt} = (G - Q_{conv} - Q_{Rad} - P)/C_{PV} - Q_{cond(CF)}/C_{p(CF)} \quad (4)$$

where C_{PV} is PV heat capacity, $C_{p(CF)}$ is the cooling fluid heat capacity, and P is the electric power generation, net solar energy incidence (G) on the PV panel, as given below:

$$G = \alpha A I_T$$

where I_T is the total solar irradiation (W/m^2), A is the PV panel area, and α is the absorptivity of the PV panel.

a) Net radiative heat transfer from the PV panel

$$Q_{Rad} = 2Ah_r(T_{PV}-T_\infty) \quad (4.1)$$

where T_∞ is the ambient temperature and h_r is radiative heat transfer coefficient is taken from (Bhakre, 2022)

$$h_r=\sigma\varepsilon(T_{PV}^2+T_{sky}^2)(T_{PV}+T_{sky})$$

where σ is Stefan Boltzmann constant (5.67×10^{-8} $Wm^{-2}K^{-4}$) and ε is the emissivity of the panel, where T_{sky} is sky temperature

$$T_{sky}= 0.037536T_\infty^{1.5}+0.32T_\infty \quad (4.2)$$

b) Heat loss from the PV panel to the ambient due to convection is defined as

$$Q_{conv}= h_c\, A\, (T_{PV} - T_\infty) \quad (4.3)$$

$$h_c = 5.7+3.8v \quad (4.4)$$

where h_c is the heat transfer coefficient, and v is wind velocity taken from the reference (Bhakre et al., 2022)

c) Heat transfer due to conduction from the PV panel to the cooling fluid can be written as

$$Q_{cond}=KA\,(T_{pv}-T_\infty)/x \quad (4.5)$$

Where k is the thermal conductivity of cooling fluid and x is the thickness of the container containing the cooling fluid.

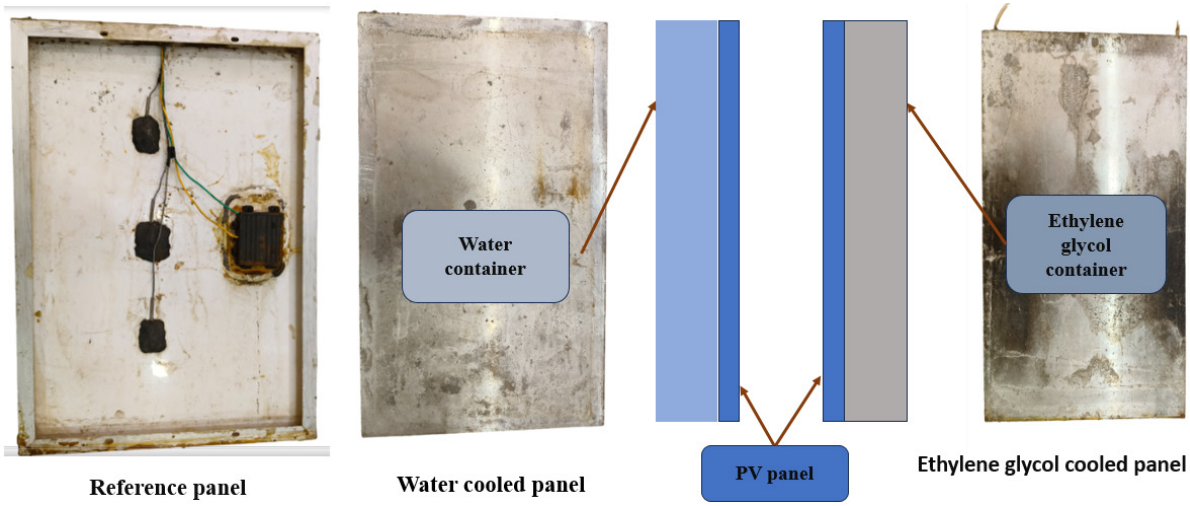


Figure 2. Developed PV panels for study purposes.

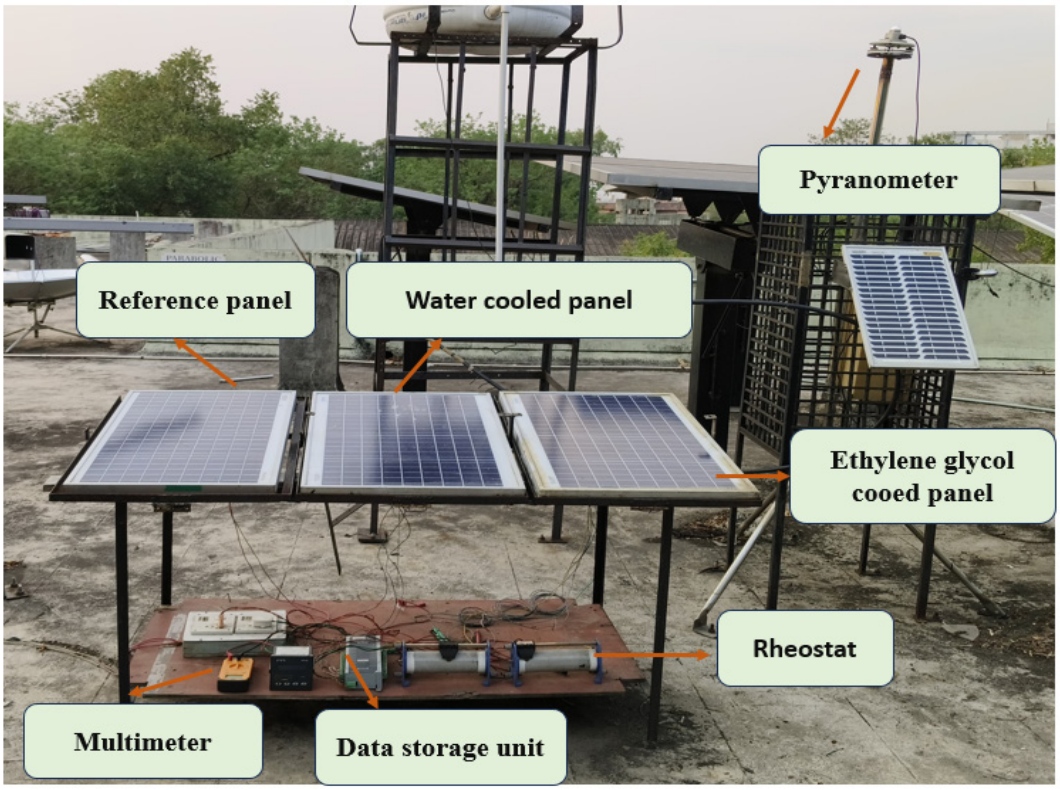


Figure 3. Actual image of the setup for experimentation.

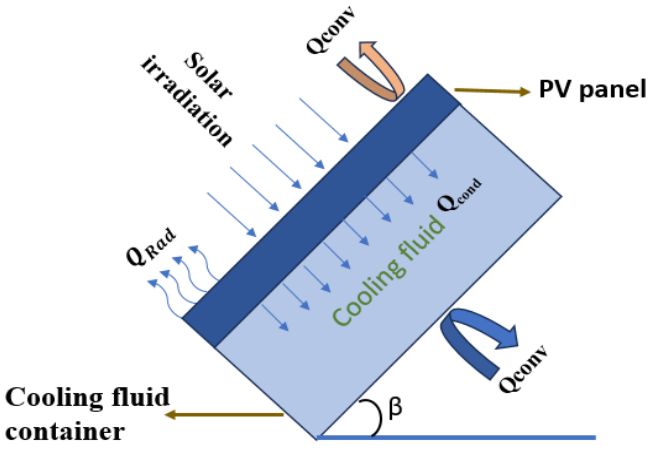


Figure 4. Model of heat transfer for the cooling system

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4. RESULTS AND DISCUSSION

The performance of the cooling panel (cooled with water and with 10% ethylene glycol mixed with water) is compared with that of the reference PV panel, and each experiment is conducted outdoors during three consecutive days of clear, sunny weather. Out of these three consecutive readings, one is taken for analysis. The analysis has focused on key performance metrics such as PV temperature, temperature reduction, electrical efficiency, and power output of the panels.

The solar radiation variation with time of day, as shown for 1, 15, and 19 March, can be seen in Figure 5. It can be seen that the trends for all three days are almost the same for solar radiation during March. In the morning, solar radiation is lower, and as noon approaches, it becomes more intense (most of the sun's energy reaches the PV panel), then starts to decrease after reaching its peak and decreases further at the end of the day. The peak solar radiation occurs between 11 AM and 12:00 noon. Maximum solar radiation was 861 W/m² at 11:27 AM on 1 March and 873 W/m² at 10:53 AM on 15 and 19 March 2024. The wind speed variation is shown in Figure 6.

4.1 Solar panel temperature profile

Figures 7 and 8 show the temperature variation of photovoltaic (PV) panels equipped with different cooling methods compared to an uncooled reference panel over the course of a day. Both figures show that the reference panel consistently reaches higher temperatures, peaking around 70°C during midday due to increased solar radiation, while the cooled panels demonstrate significantly lower temperatures.

In Figure 7, the water-cooled panel exhibits a slower rise in temperature compared to the reference panel, with a peak temperature of approximately 60°C. This demonstrates the ability of water cooling to mitigate thermal accumulation, thereby enhancing the panel's thermal regulation. Figure 8 presents the performance of a PV panel cooled with a 10% ethylene glycol mixed with water (EGMWW) solution. Compared to water cooling, the EGMWW-cooled panel achieves even greater thermal regulation, maintaining a lower peak temperature well below 60°C.

After 2:30 PM, the temperature of the non-cooled panel is lower than that of the cooled panel, which utilizes a water and 10% ethylene glycol mixture. The ethylene glycol enhances PV panel performance by allowing for rapid heat dissipation. The temperature difference between the non-cooled panel and the 10% ethylene glycol mixed with water cooled panel is less than that between the non-cooled panel and the panel that is only water-cooled.

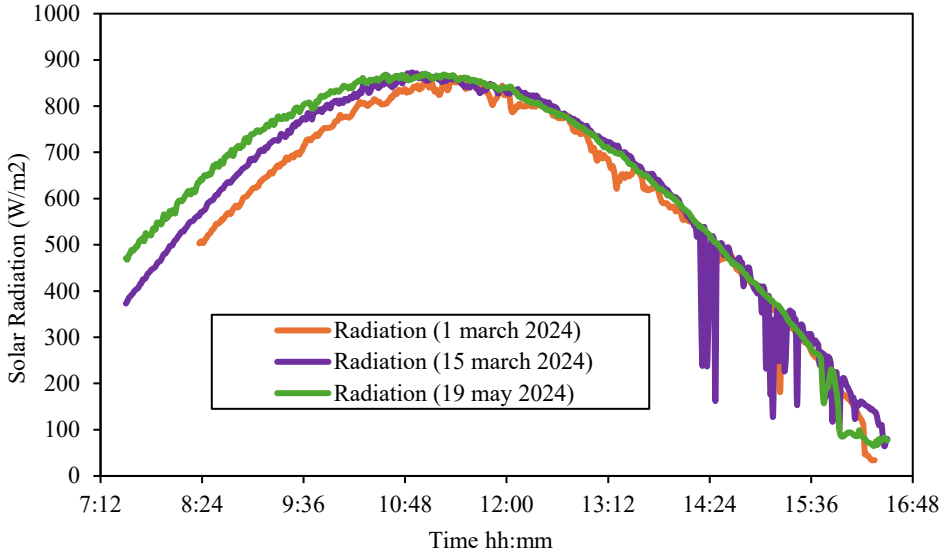


Figure 5. Variation of solar radiation with respect to time

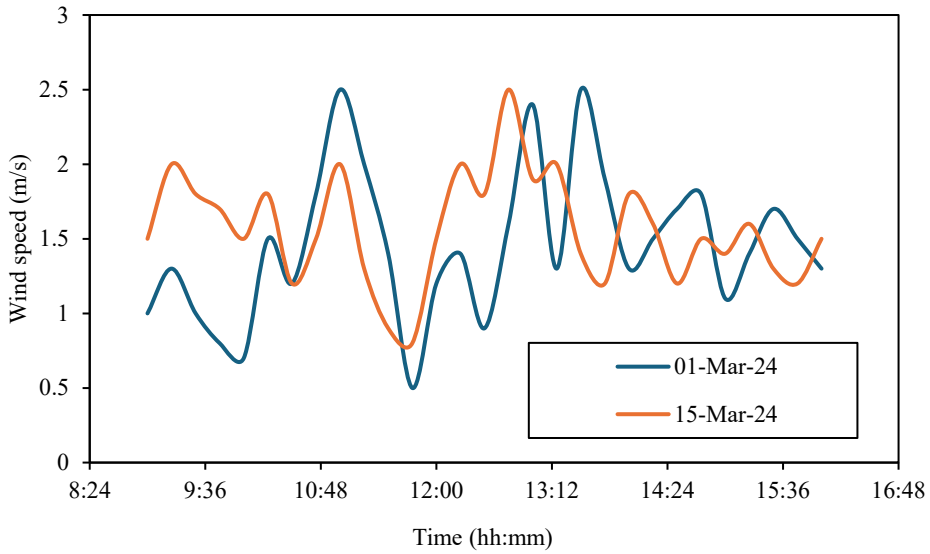


Figure 6. Wind speed variation with time over the experiment days.

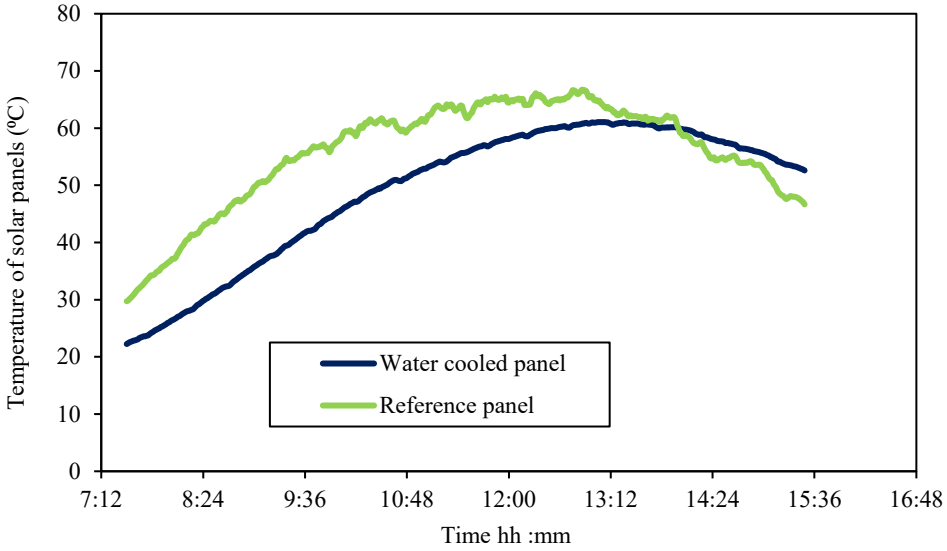


Figure 7. Variation in the temperature of the water-cooled panel and reference panel on March 1, 2024.

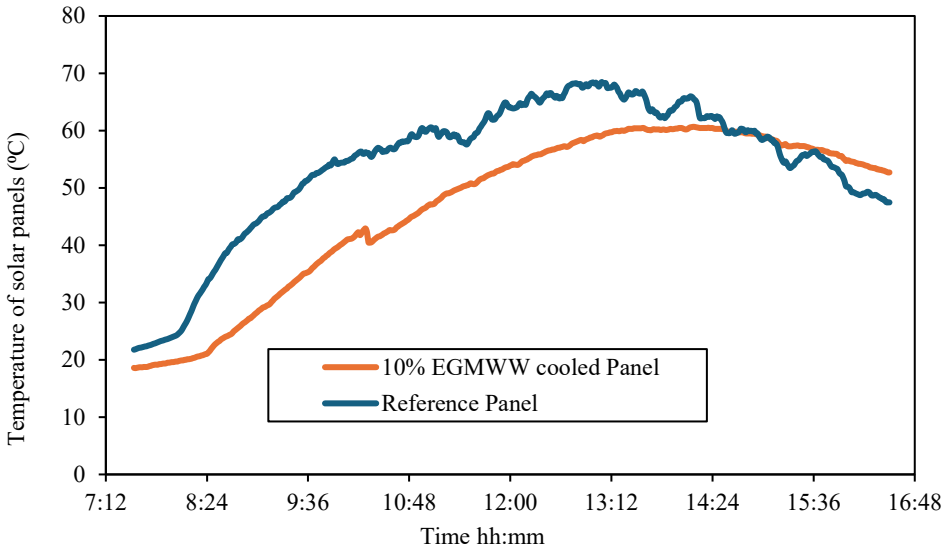


Figure 8. Variation in temperature of the 10% EGMWW cooled panel and reference panel on March 15, 2024.

At 13:44, the water-cooled PV system achieved its peak temperature, after which it dropped. The maximum temperatures observed for the non-cooled panel and the water-cooled panel were 66.73°C and 60.8°C, respectively. The ethylene glycol-cooled system reached its maximum temperature at 13:01, with the highest temperatures recorded for the non-cooled and ethylene *glycol*-cooled panels being 68.36°C and 59.1°C, respectively. Both the ethylene glycol- and water-cooled systems attained the same temperature as the non-cooled panel at 14:30. Up to 13:30, the temperature of the cooled panel remained lower than that of the non-cooled panel. The average temperatures of the non-cooled panel, water-cooled panel, and 10% EGMWW panel are 66.73°C, 60.8°C, and 59.1°C, respectively. This demonstrates a temperature drop of 8.88% for the water-cooled panel and 13.55% for the ethylene glycol-cooled panel.

4.2 Power output profile of Solar PV panels

When the applied load is taken into account, the maximum power output (P_{max}) of the photovoltaic panel is calculated using Equation (5). This is defined as the product of the maximum voltage (V_{max}) and maximum current (I_{max}).

$$P_{max}=V_{max}\times I_{max} \quad (5)$$

Figures 9 (a) and (b) show the power output profiles of photovoltaic (PV) panels with different cooling methods compared to an uncooled reference panel throughout the day. In Figure 9 (a), the reference panel exhibits a maximum power output of 27 W and an average output of 22 W, while the water-cooled panel achieves a higher maximum of 30 W and an average of 25 W, indicating a performance improvement of approximately 3 W. In Figure 9 (b), the panel cooled with a 10% ethylene glycol mixed with water (EGMWW) solution outperforms both, with a maximum power output of 32 W and an average of 28 W, representing increases of 5 W and 6 W, respectively, compared to the reference panel. These results highlight the effectiveness of thermal management in enhancing PV panel efficiency, with the EGMWW solution demonstrating superior performance over water cooling due to its better thermal conductivity and heat dissipation capabilities, resulting in improved energy yield and operational efficiency. Ethylene glycol works better than water as a coolant because it has higher viscosity, lower vapor pressure, and better thermal stability. These factors improve coolant stability and reduce evaporation losses at high temperatures. While water has higher

thermal conductivity and heat capacity, adding 10% ethylene glycol improves flow stability, prevents particle clumping, and keeps the temperature consistent across the PV surface. This balance between heat transfer and stability makes the water and ethylene glycol mixture more effective for long-lasting PV/T cooling performance in hot and dry climates.

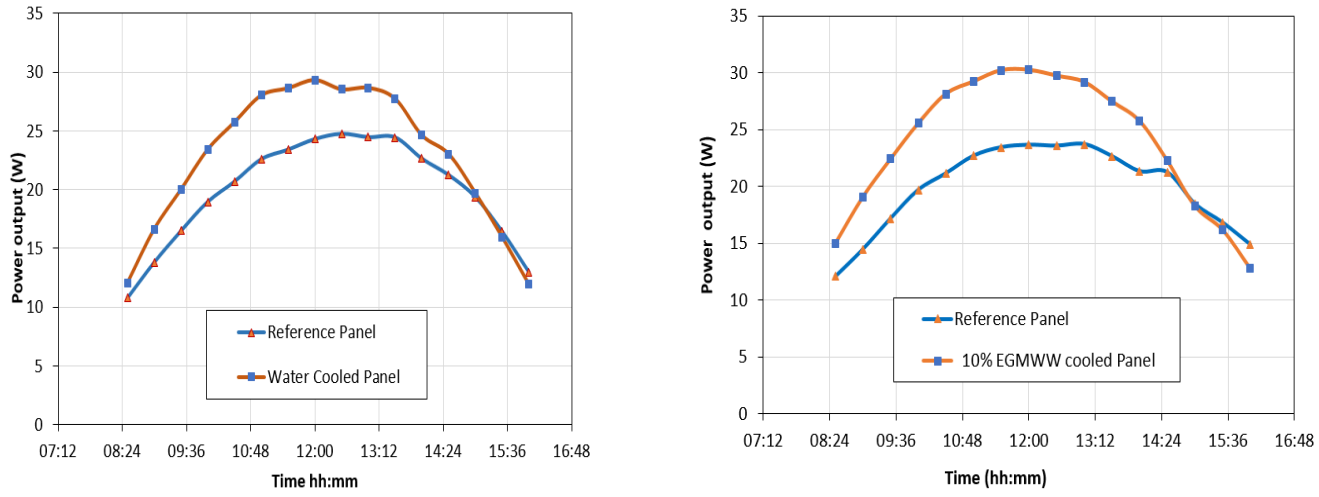
4.3 Electrical efficiency profile of Solar PV panel

Electrical efficiency is used to measure the performance of photovoltaic panels. PV cell temperature has a significant impact on PV panel efficiency. Equation (6) proposed by Chandrasekar et al. (2013) is used to calculate the efficiency of a photovoltaic panel:

$$\eta_{PV} = \eta_{ref}[1 - \beta_{ref} (T_{PV-} T_{ref})] \quad (6)$$

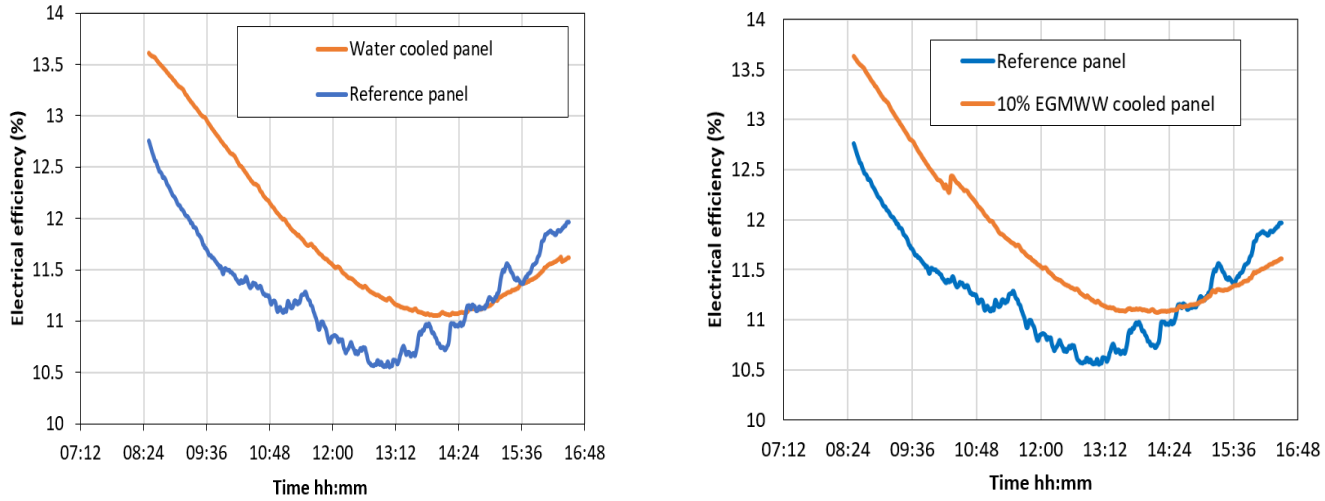
where η_{PV} represents the efficiency of solar and T_{PV} is the temperature at which photovoltaic cells are operating. At a reference temperature, η_{ref} represents the reference efficiency.

At a reference temperature T_{ref}, β_{ref} represents the temperature coefficient. The manufacturer has defined values for η_{PV}, β_{ref}, and T_{ref}, which are 0.1348, 0.005/ºC, and 25ºC, respectively. These very closely resemble the values provided by Chandrasekar & Senthilkumar (2015) for crystalline silicon panels. Using water and 10% ethylene glycol mixed with water results in a considerable decrease in PV panel temperature, which enhances electrical conversion efficiency. The electrical efficiency variation of the non-cooled panel, water-cooled panel, and 10% ethylene glycol mixed with water (10% EGMWW) panel during the experiments is shown in Figures 10 (a) and (b). As can be seen from the figures, the cooling system significantly improved panel efficiency. As can be observed, the water-cooled system and the 10% EGMWW system exhibit greater electrical efficiency values than the reference panel. Over the course of the day, the water-cooled and 10% EGMWW systems had average electrical efficiencies of 11.79% and 11.82%, respectively, compared to 11.28% for the non-cooled PV panel. This indicates an average increase in efficiency of 4.48% and 4.72% over the non-cooled reference PV panel. During the periods of 9:55 AM and 9:38 AM, the maximum increases in electrical efficiency compared to the reference PV system were 9.2% and 11.0% for the water-cooled and 10% EGMWW systems, respectively.



(a) (b)

Figure 9. Time-varying variations in PV panels' power output (a) water cooled (1 March 2024) (b) 10% ethylene glycol mixed with water (15 March 2024).



(a) (b)

Figure 10. Time-varying variations in PV panels' electrical efficiency (a) water cooled (1 March 2024) (b) 10% ethylene glycol mixed with water (15 March 2024).

5. CONCLUSIONS

A passive cooling system integrating water-cooled and 10% ethylene glycol mixed water-cooled systems was investigated in real outdoor conditions. The following conclusions are drawn from the study:

- The experimental results revealed that both cooling methods significantly enhanced power generation and electrical efficiency compared to the reference PV panel.
- The 10% ethylene glycol mixed water-cooled system achieved slightly higher performance than the water-cooled system.
- The maximum temperatures recorded for the reference panel, water-cooled panel, and 10% ethylene glycol mixed water-cooled panel were 66.73 °C, 60.8 °C, and 59.1 °C, respectively.
- Power generation increased by 5.04 W for the water-cooled system and 6.603 W for the glycol-water system, with average temperature reductions of 9.2% and 11.0%, leading to electrical efficiency improvements of 4.48% and 4.72%, respectively. To optimize performance, changing the cooling fluid daily around midday is recommended.
- The present study's findings highlight better performance in terms of PV panel output. Future work can be carried out by considering the flow of the mixture at different mass flow rates.

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